5.2 Colour schemes and types of data

Watch: Purposeful use of colour – part 2 [12.01 mins]

Watch the following video, which explains what a colour scheme is and how colour schemes can be set and used.

This is useful as it enhances skills regarding the use of colour.

0:00/12:01

[Transcript (https://learn.usc.edu.au/courses/35242/files/3138644?wrap=1)](https://learn.usc.edu.au/courses/35242/files/3138644?wrap=1)

Source: © The University of Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Purposeful use of colour – part 2 (https://learn.usc.edu.au/courses/35242/files/3138662?wrap=1)](https://learn.usc.edu.au/courses/35242/files/3138662?wrap=1)

Considerations when making decisions about colour

There are a number of considerations to take into account when deciding on what colour scheme to use in your data visualisation.

Colour scheme

This is a set of colours that are to be used together in one or a series of data visualisations.

Categorical colour schemes

Categorical variables represent discrete groups. In general, visualisations of categorical variables communicate information about the absolute or relative frequency of the group/s. Where there is no inherent order, it is useful to represent each group through the use of a distinct colour. The result is a categorical colour scheme. The use of distinct colours is important, as it improves readability for the audience. However, categorical colour schemes should be limited to no more than six colours. This is because it is challenging for audience members to distinguish between groups through the colour associated with them.

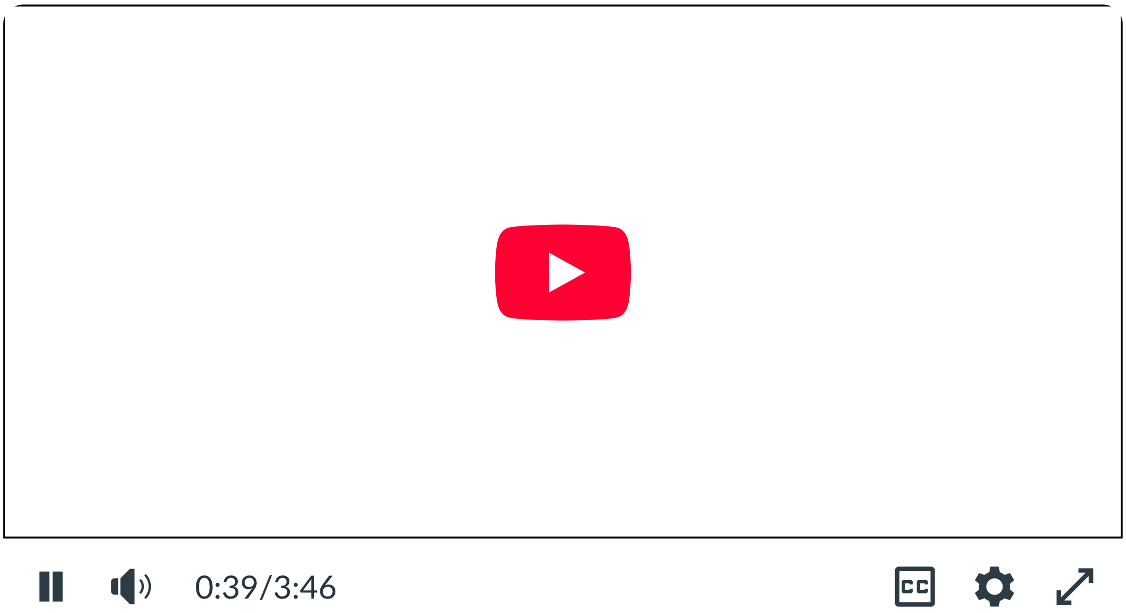
Sequential colour schemes

A sequential colour scheme relates to a scheme that utilises the relative degree of saturation or luminance of a hue to create a gradient that represents the relative value of the variable. Sequential colour schemes are useful when the values of a variable can be arranged in ascending or descending order. In sequential colour schemes, it is more common to use degrees of luminance rather than degrees of saturation.

Diverging colour schemes

A diverging colour scheme uses a gradient of colour. It is formed by the use of two sequential colour schemes which share an end point at the reference value. Two hues are used. One is associated with values that are below the reference value. The other is associated with values above the reference value. The luminance of the hue communicates the direction of deviation from the reference point. Typically, distinctive hues are used. Primary hues make it easier to distinguish direction and the degree of deviation. Diverging colour schemes are used most effectively when there is a need to highlight both extremes of the value of a variable.

> ### Watch: How to use colour in your data visualisation – beginner’s guide [3.46 mins]
>
> This video is useful, as it explains how to use the four principles of colour palette structures in data visualisations.
>
> As you watch consider:
>
> - How does understanding these four principles help to improve your data storytelling skills?
> - Why is colour and the careful use of it important?

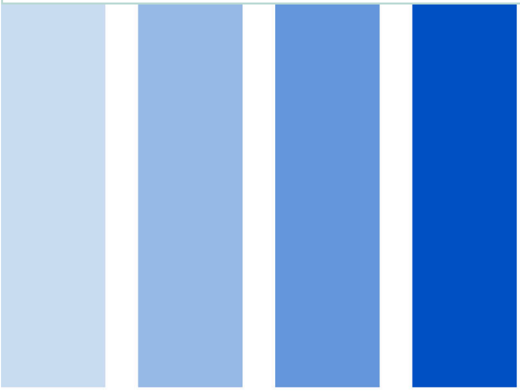


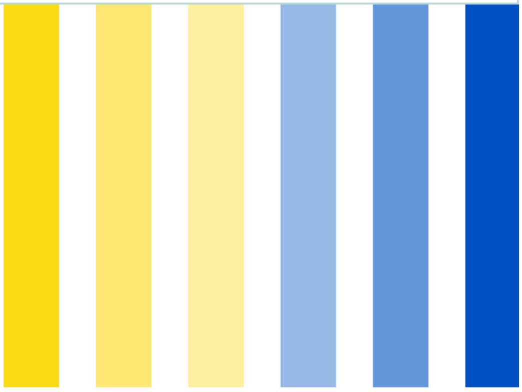
Source: © Undatable (2021).

Activity: Check your understanding – principles of colour palettes

Consider each image below and label them by dragging the title to the correct image.

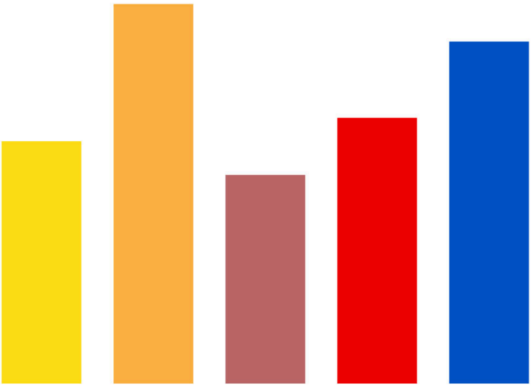
Attempt successfully submitted

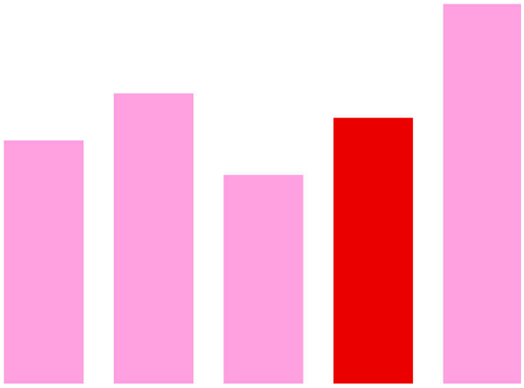




Sequential

Diverging





Categorical

Highlight

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